# ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** magnetic | Magnetic materials include iron, nickel and cobalt. |
| 2 | **(a)** small magnet | It is a tiny magnet. |
| 3 | **(a)** to keep them from losing magnetism | Keepers preserve magnetism. |
| 4 | **(d)** iron nail | Iron is magnetic. |
| 5 | **(d)** north-south | It aligns with Earth's field. |
| 6 | **(a)** mobile phones and credit cards | They may damage magnetic data. |
| 7 | **(d)** glass marble | Glass is not attracted. |
| 8 | **(c)** horseshoe magnet | U shaped. |
| 9 | **(a)** sailors | They find directions at sea. |
| 10 | **(d)** poles | Poles are at the ends. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Magnetic: iron, nickel, cobalt (1); Non-magnetic: wood, plastic, glass (1). | 1+1 |
| 12 | It is influenced by the Earth's magnetism (1); The Earth behaves like a large magnet (1). | 1+1 |
| 13 | They keep the magnetism from fading (1); They complete the magnetic pairs (1). | 1+1 |
| 14 | The ends where attraction is strongest (1); Called north and south poles (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Move a magnet over the mixture (1); Iron nails stick to the magnet (1); Wooden pieces remain (1). | 1+1+1 |
| 16 | Keep in pairs with keepers (1); Avoid heating and hammering (1); Keep away from electronic gadgets (1). | 1+1+1 |
| 17 | Sprinkle filings around the magnet (1); They line up in curved lines (1); Lines are densest near the poles (1). | 1+1+1 |
| | **OR:** Bring the magnet near the compass (1); Note which end attracts the north of the needle (1); That end is the south pole of the magnet (1). | 1+1+1 |
| 18 | Magnetic: attracted, e.g. iron (1); Non-magnetic: not attracted, e.g. plastic (1); Test with a magnet (1). | 1+1+1 |
| | **OR:** The nearby magnet attracts the needle (1); Its force is stronger than Earth's (1); Needle is deflected (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Collect various objects (1); Bring a magnet near each (1); Magnetic: iron nail, steel pin (1); Non-magnetic: wood, plastic, glass (1); Record in a table (1). | 1+1+1+1+1 |
| | **OR:** Magnetised needle on a pivot (1); Dial with directions (1); Hold it flat (1); Needle points north (1); Rotate the dial to match N (1). | 1+1+1+1+1 |
| 20 | Earth behaves like a bar magnet (1); A freely suspended magnet aligns north-south (1); Compass needle is a magnet (1); It settles in the north-south direction (1); Directions are found from this (1). | 1+1+1+1+1 |
| | **OR:** Place paper on the magnet (1); Sprinkle iron filings (1); Tap the paper (1); Filings arrange in curves (1); They crowd near the poles (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# THE MASTERMIND COACHING CLASSES

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(c)** bar | Bar magnets are common. |
| 2 | **(d)** it repels a known magnet | Only a magnet can repel another magnet. |
| 3 | **(d)** north and south | North-seeking and south-seeking poles. |
| 4 | **(c)** has two poles | Each piece is a magnet with N and S poles. |
| 5 | **(b)** always | Magnetic force attracts iron. |
| 6 | **(c)** north-south direction | It aligns with Earth's magnetic field. |
| 7 | **(c)** heated or hammered | Heating and hammering weaken them. |
| 8 | **(c)** the poles | Poles attract most iron filings. |
| 9 | **(a)** north pole | It is the north-seeking pole. |
| 10 | **(d)** rubbing with a magnet | Stroking with a magnet in one direction. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Bar (1); Horseshoe or ring (1). | 1+1 |
| 12 | Each piece becomes a magnet (1); Each has a north and a south pole (1). | 1+1 |
| 13 | Like poles repel (1); Unlike poles attract (1). | 1+1 |
| 14 | Stroke it with a magnet in one direction (1); Repeat many times (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Hang a bar magnet (1); Bring the same pole of another magnet near (1); It moves away, showing repulsion (1). | 1+1+1 |
| 16 | They attract (1); They may stick together (1); Unlike poles attract (1). | 1+1+1 |
| 17 | Refrigerator door (1); Pencil box closer (1); Speaker (1). | 1+1+1 |
| | **OR:** Iron, nickel, cobalt (1); Steel is used in tools (1); Magnets are used in speakers (1). | 1+1+1 |
| 18 | It is a small magnet (1); It turns freely (1); It settles in the north-south direction (1). | 1+1+1 |
| | **OR:** A force by a magnet (1); Acts without touching (1); Can attract or repel (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Take a steel bar (1); Stroke with one pole of a magnet in one direction (1); Lift magnet at the end, repeat many times (1); Test using iron pins (1); The bar picks up pins (1). | 1+1+1+1+1 |
| | **OR:** Natural magnet called lodestone (1); It attracted iron (1); It pointed north-south when hung (1); Used for navigation (1); Led to the compass (1). | 1+1+1+1+1 |
| 20 | Suspend one bar magnet (1); Bring N pole of another near N pole: repulsion (1); Bring S pole near N pole: attraction (1); Like repel, unlike attract (1); Repulsion confirms a magnet (1). | 1+1+1+1+1 |
| | **OR:** Iron from sand (1); Iron from waste (1); Steel from scrap (1); Magnet picks only magnetic items (1); Others are left behind (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

# ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

## SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(d)** the ends | They cluster at the poles. |
| 2 | **(c)** repel each other | North repels north. |
| 3 | **(d)** iron, nickel, cobalt | These are the main magnetic metals. |
| 4 | **(d)** lodestone | Lodestone is a natural magnet. |
| 5 | **(c)** attract each other | North attracts south. |
| 6 | **(a)** aluminium foil | Aluminium is not attracted. |
| 7 | **(d)** it can act from a distance | Non-contact force. |
| 8 | **(b)** magnetic compass | A compass has a magnetised needle. |
| 9 | **(d)** magnetic force acts through paper | Magnetic force acts across non-magnetic materials. |
| 10 | **(d)** steel paper clip | Steel contains iron. |

## SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Compass (1); Door closers or fridge doors (1). | 1+1 |
| 12 | Its needle always points north-south (1); We find other directions from this (1). | 1+1 |
| 13 | Heating it (1); Hammering or dropping it (1). | 1+1 |
| 14 | They may damage the phone (1); They disturb its compass or memory (1). | 1+1 |

## SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | No landmarks to guide (1); Needle gives north-south (1); Helps in finding direction (1). | 1+1+1 |
| 16 | Magnet repels one end of another magnet (1); Iron nail is only attracted (1); Repulsion shows a magnet (1). | 1+1+1 |
| 17 | Bring a magnet near (1); If attracted, magnetic (1); Else not (1). | 1+1+1 |
| | **OR:** Take a bar magnet (1); Stroke needle from end to end in one direction (1); Test by picking up pins (1). | 1+1+1 |
| 18 | Place paper between magnet and pin (1); Pin is still attracted (1); Force passes through non-magnetic sheets (1). | 1+1+1 |
| | **OR:** Hang the magnet freely (1); The end that points north is north pole (1); The other end is south pole (1). | 1+1+1 |

## SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Heating (1); Hammering (1); Dropping (1); Keep in pairs (1); Place keepers (1). | 1+1+1+1+1 |
| | **OR:** Each piece is a magnet (1); Each has N and S pole (1); Smaller pieces remain magnets (1); Poles cannot be isolated (1); Strength decreases (1). | 1+1+1+1+1 |
| 20 | Compass (1); Fridge door (1); Loudspeaker (1); Cranes lifting scrap iron (1); Magnetic toys (1). | 1+1+1+1+1 |
| | **OR:** Attracts magnetic materials (1); Has two poles (1); Like poles repel (1); Unlike poles attract (1); Points north-south when free (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.